Activity 1, Intermediate - Code & Go Mouse

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In this adventure, students will learn coding skills with Colby, the programmable robot mouse from APH. The students will program the sequence of steps and Colby will race to find the cheese.

Setup - Activity 1

This activity can take several sessions. Students will need time to create the maze before solving it.

Objectives

- Students will be able to learn basic coding skills.
- Students will be able to understand the basic components of computational thinking.

Materials Needed

- APH's Code & Go Robot Mouse
- Assorted textured paper
- Stickers
- Printable 5" x 5" squares
- Scissors
- Glue Stick
- Other art supplies

Ideas for Carrying Out This Activity

- Work in some route planning and O&M skills to your adventure.
- If students are having trouble identifying right and left according to the mouse's perspective. Try recreating the grid with foam mat floor tiles and have the student pretend to be Colby. The student can walk through the life-size course to get a better understanding of the mouse's perspective.

- Allowing students to arrange these tiles to create their own grid, shape or "road" encourages the student to be creative. Taking the tactile diagram and "building" the diagram is another form of creativity. An advanced student could take this a step farther by creating their own Code and Go course then create their version of the accompanying Activity Card using graph paper and tactile materials.

Adventure Map: Activity 1

Teaching tip: Provide sufficient time for the student to explore, develop skills, and have fun at each step! Encouraging creativity and personal preferences for drawing as much as possible. Some students may be able to accomplish each step in one session; most students will need several sessions to complete the adventure.

1. Introduction

- Ask your student: How does developing a route for Colby(the mouse) relate to writing code?
  - Review vocabulary with your student:
  - Algorithmic Thinking - a way of approaching problems that involves breaking them down into smaller, more manageable parts. It is a process that involves identifying the steps needed to solve a problem and then implementing those steps in a logical and efficient manner.
  - Debug - identify and remove errors from a computer program.

2. Introduce Code & Go

- Let the student explore all the components of the Code & Go set.
- Let them figure out how the tiles fit together.
- Let them try to control Colby the mouse before telling them how it works.
- Let them guess the purpose of the other parts.

3. Practice Together

- Demonstrate how the robot mouse, Colby, can be programmed by using the buttons.
- Set up a simple maze, show the student where Colby should start and where they should end up (at the cheese).
- Have the student plan out Colby's route using the arrow cards.
- Have the student program Colby and see if their program works.
- If something goes wrong, have the student try to debug their program, go back through their initial plan with the arrow cards and see if they can figure out the step where things went wrong.
- After the maze has been solved, have the student try another, harder maze.

- You can use the maze walls to create traps and dead-ends and bridges to guide Colby in a certain direction.

4. **Design Colby's Mouse Town**
   - To give your student additional opportunities to explore the idea of creating a town for Colby and be creative. Use the 5" x 5" square (this is the size of the tiles from Code and Go) and have your student draw out/ design the places in the town. Some examples are a School, Library, Park and Cheese Shop. Make sure to reiterate sketching techniques covered from the introductory adventures. They can use markers, stickers, textured paper, braille labels or any other art supplies you have.

5. **Sharing and Discussion**
   - What did we learn today?
   - How is giving Colby directions like writing a computer program?
   - How important do you think debugging is in programming?

6. **Conclusion**
   - Summarize the key points of the adventure.
   - Have the student reflect on what went well and what was challenging.
   - If time allows, have the students create their own mazes.

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